

161 $n=10000$

$$a) p(|\xi - 0.5| \geq 0.01) \leq \frac{pq}{n\varepsilon^2} = \frac{1/2 \cdot 1/2}{10^4 \cdot (0.01)^2} = \frac{1}{4} = 0.25$$

$$b) p(|\xi - 0.5| \geq 0.01) = 1 - 2\Phi\left(\frac{0.1 \cdot \sqrt{10^4}}{\sqrt{\frac{1}{2} \cdot \frac{1}{2}}}\right) = 1 - 2\Phi(2) = 1 - 0.9544 = 0.0456$$

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$$p\left(\left|\frac{S_n}{n} - \frac{1}{\alpha}\right| < \frac{1}{\alpha}\right) = 1 - p\left(\left|\frac{S_n}{n} - E\xi\right| \geq \frac{1}{\alpha}\right) \geq 1 - \frac{1/\alpha^2}{10 \cdot \frac{4}{\alpha^2}} = 1 - \frac{1}{40} = \frac{39}{40}$$

$$n=10 \quad \frac{S_n}{n} \in E_\alpha \quad E\xi_i = \frac{1}{\alpha} \quad D\xi_i = \frac{1}{\alpha^2}$$

169 $\xi \in N(m, 1) \quad \frac{S_n}{n} \in N(m, \frac{1}{16}) \quad n=16$

$$p(\xi > 5) = p\left(\frac{S_n}{n} < 1\right)$$

$$p(5 < \xi < +\infty) = p(-\infty < \frac{S_n}{n} < 1)$$

$$\Phi(+\infty) - \Phi(5-m) = \Phi\left(\frac{1-m}{1/\sqrt{n}}\right) - \Phi(-\infty)$$

$$\frac{1}{2} - \Phi(5-m) = \Phi(4(1-m)) + \frac{1}{2}$$

$$\Phi(m-5) = \Phi(4-4m) \Rightarrow m-5 = 4-4m \Rightarrow m = 1.8 \quad \text{Ombet: } m=1.8$$